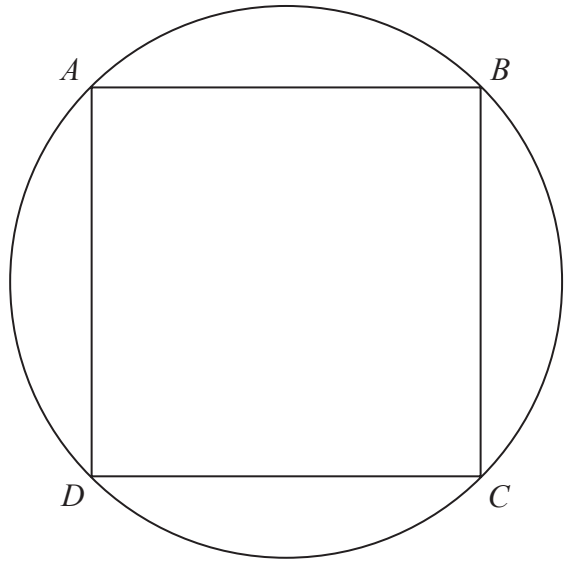
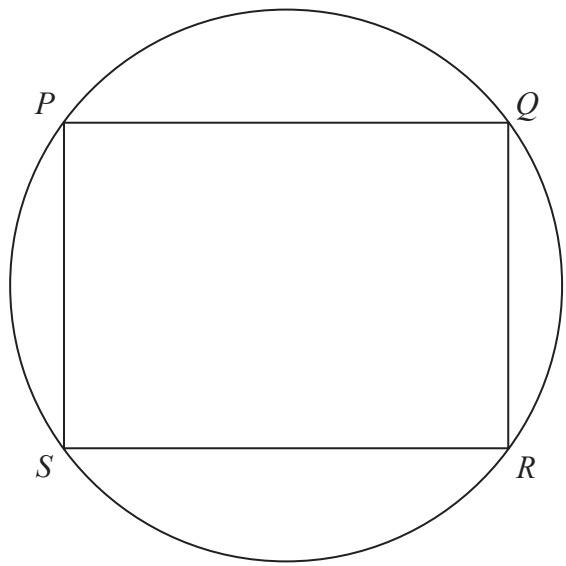
24 *ABCD* is a square with sides 10 cm.
The vertices of the square lie on a circle, $\mathbf{C_1}$



PQRS is a rectangle with sides $(10 + x)$ cm and $(10 - x)$ cm where $0 < x < 10$
The vertices of the rectangle lie on a circle, $\mathbf{C_2}$



Prove that the area of $\mathbf{C_2}$ is greater than the area of $\mathbf{C_1}$

(Total for Question 24 is 4 marks)